higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MATHEMATICS N6

18 NOVEMBER 2019

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QUESTION 1

1.1 1.1.1

$$z = \ln\ (\sqrt{x} + \sqrt{y}\ )^2$$

$$= 2\ln\ (\sqrt{x} + \sqrt{y}\ )$$

$$\frac{\partial z}{\partial x} = \frac{2}{\sqrt{x}+\sqrt{y}}\frac{1}{2\sqrt{x}}\ \checkmark\checkmark$$

$$= \frac{1}{\sqrt{x}(\sqrt{x}+\sqrt{y})}.$$

or $\frac{\partial z}{\partial x} = \frac{1}{\left(\sqrt{x}+\sqrt{y}\right)^2} 2\left(\sqrt{x}+\sqrt{y}\right)\frac{1}{2\sqrt{x}}$ √ √

$= \frac{1}{\sqrt{x}\left(\sqrt{x}+\sqrt{y}\right)}$

OR

$$z = \ln\left(\sqrt{x}+\sqrt{y}\right)^2$$

$$= \ln\left(x + y + 2\sqrt{x}\sqrt{y}\right)$$

$$\frac{\partial z}{\partial x} = \frac{1}{x+y+2\sqrt{x}\sqrt{y}}\left(1+\frac{2\sqrt{y}}{2\sqrt{x}}\right)\surd\surd$$

$$= \frac{1}{x+y+2\sqrt{x}\sqrt{y}}\left(\frac{\sqrt{x}+\sqrt{y}}{\sqrt{x}}\right)$$

$$= \frac{1}{\left(\sqrt{x}+\sqrt{y}\right)^2}\left(\frac{\sqrt{x}+\sqrt{y}}{\sqrt{x}}\right)$$

$$= \frac{1}{\sqrt{x}\left(\sqrt{x}+\sqrt{y}\right)}$$

(1)

1.1.2

$$\frac{\partial^2 z}{\partial y \partial x} = \frac{1}{\sqrt{x}}\frac{-1}{(\sqrt{x}+\sqrt{y})^2}\frac{1}{2\sqrt{y}}\quad \surd\surd\surd\surd$$

$$= \frac{-1}{2\sqrt{x}\sqrt{y}\,(\sqrt{x}+\sqrt{y})^2}$$

Or $\frac{\partial^2 y}{\partial y\,\partial x} = \frac{1}{\sqrt{x}}\left[\frac{\left(\sqrt{x}+\sqrt{y}\right)\times 0 - 1\left(0+\frac{1}{2\sqrt{y}}\right)}{\left(\sqrt{x}+\sqrt{y}\right)^2}\right]$

√√√√

(2)

1.2 1.2.1

$x = \sec 2\theta \qquad \frac{dx}{d\theta} = 2\sec 2\theta \tan 2\theta \quad \surd$

$y = \tan 2\theta \qquad \frac{dy}{d\theta} = 2\sec^2 2\theta \quad \surd$

$\frac{dy}{dx} = 2\sec^2 2\theta \frac{1}{2\sec 2\theta \tan 2\theta} \quad \surd$

$= \frac{\sec 2\theta}{\tan 2\theta}$

$= \frac{1}{\cos 2\theta}\frac{\cos 2\theta}{\sin 2\theta}$

$= \frac{1}{\sin 2\theta}$ or $\cos ec2\theta \quad \surd$ (2)

1.2.2

$\frac{d^2y}{dx^2} = \frac{d}{d\theta}\left(\frac{dy}{dx}\right)\frac{d\theta}{dx}$

$= -2\cos ec2\theta \cot 2\theta \frac{1}{2\sec 2\theta \tan 2\theta}$

$= -\cot^3 2\theta$ or $-\frac{1}{\tan^3 2\theta}$ ✓✓

Or $\frac{d^2y}{dx^2} = \frac{d}{d\theta}\left(\frac{sec\,2\theta}{\tan 2\theta}\right) \times \frac{1}{2\,sec\,2\theta tan\,2\theta}$

¿ $\frac{\tan 2\theta \times 2\,sec\,2\theta tan\,2\theta - sec\,2\theta \times 2\,sec^2 2\theta}{\tan^2 2\theta} \times \frac{1}{2\,sec\,2\theta tan\,2\theta}$ √√

, $2\,sec\,2\theta\left(\tan^2 2\theta - sec^2 2\theta\right)$ (1)

[6]

QUESTION 2

2.1

$$\int y\,dx$$

$$= \int \sin^3(2x+3)\,dx$$

$$= \int \sin^2(2x+3)\sin(2x+3)\,dx \quad \surd$$

$$= \int \left(1-\cos^2(2x+3)\right)\sin(2x+3)\,dx \quad \surd$$

$$= \int \sin(2x+3)\,dx - \int \cos^2(2x+3)\sin(2x+3)\,dx \quad \surd$$

$$= -\frac{1}{2}\cos(2x+3) + \frac{1}{2}\frac{1}{3}\cos^3(2x+3) \quad \surd\surd\surd$$

or using $u = \cos(2x+3)$

$$\frac{du}{dx} = -2\sin(2x+3)$$

$$dx = \frac{du}{-2\sin(2x+3)}$$

(3)

2.2

$$2\left(x^2 + \frac{5}{2}x + 3\right)$$

$$= 2\left(x^2 + \frac{5}{2}x + \left(\frac{5}{4}\right)^2 + 3 - \left(\frac{5}{4}\right)^2\right) \surd\surd$$

$$= 2\left[\left(x+\frac{5}{4}\right)^2 + \frac{23}{16}\right] \surd\surd$$

$$\int y\,dx = \int \frac{1}{2\left(x^2 + \frac{5}{2}x + 3\right)}\,dx$$

$$= \int \frac{1}{2\left[\left(x+\frac{5}{4}\right)^2 + \frac{23}{16}\right]}\,dx \quad \surd$$

$$= \frac{1}{2}\frac{1}{\frac{\sqrt{23}}{4}.1}\tan^{-1}\frac{x+\frac{5}{4}}{\frac{\sqrt{23}}{4}} + c \quad \surd\surd\surd$$

$$= \frac{2}{\sqrt{23}}\tan^{-1}\frac{4x+5}{\sqrt{23}} + c$$

Alternative

$$2\left(x^2 + \frac{5}{2}x + 3\right) = 2x^2 + 5x + 6$$

$$ax^2 + bx + c = \frac{4ac - b^2}{4a} + a\left(x + \frac{b}{2a}\right)^2$$

$$\therefore 2x^2 + 5x + 6 = \frac{4.2.6 - 5^2}{4.2a} + 2\left(x + \frac{5}{2.2}\right)^2$$

$$= \frac{23}{8} + 2\left(x + \frac{5}{4}\right)^2$$

$$\int \frac{1}{2\left(x^2 + \frac{5}{2}x + 3\right)}dx = \int \frac{1}{\frac{23}{8} + 2\left(x + \frac{5}{4}\right)^2}dx$$

$$\frac{1}{\frac{\sqrt{23}}{\sqrt{8}}.\sqrt{2}}\tan^{-1}\frac{\sqrt{2}(x+\frac{5}{4})}{\frac{\sqrt{23}}{\sqrt{8}}} + C$$

$$\frac{2}{\sqrt{23}}\tan^{-1}\frac{4x+5}{\sqrt{23}} + C$$

(4)

2.3

$$\int y\,dx = \int \frac{\tan^{-1}\sqrt{x}}{\sqrt{x}}dx$$

$$= \int x^{-\frac{1}{2}} \tan^{-1}\sqrt{x}\,dx \quad \surd \qquad f(x) = \tan^{-1}\sqrt{x} \qquad f'(x) = \frac{\frac{1}{2\sqrt{x}}}{1+x} = \frac{1}{2\sqrt{x}(1+x)} \quad \surd$$

$$g'(x) = x^{-\frac{1}{2}} \qquad g(x) = 2x^{\frac{1}{2}} \surd$$

$$= \tan^{-1}\sqrt{x}\ .2x^{\frac{1}{2}} - \int \frac{1}{2\sqrt{x}(1+x)} 2x^{\frac{1}{2}}dx \qquad \surd\surd\surd$$

$$= \tan^{-1}\sqrt{x}\ .2x^{\frac{1}{2}} - \int \frac{1}{1+x}dx \quad \surd$$

$$= \tan^{-1}\sqrt{x}\ .2x^{\frac{1}{2}} - \ln(1+x) + c \surd$$

Or $2\sqrt{x}\tan^{-1}\sqrt{x} - \ln(1+x) + C$

Or $\int ydx = \int \frac{1}{\sqrt{x}}\tan^{-1}\sqrt{x}\,dx \quad u = \sqrt{x} \Rightarrow \frac{du}{dx} = \frac{1}{2\sqrt{x}} \surd$

¿ $\int \frac{1}{\sqrt{x}}\tan^{-1}u \times 2\sqrt{x}\,du \surd$

$= 2\int 1 \times \tan^{-1}u\,du \quad \surd \qquad f(u) = \tan^{-1}u \Rightarrow f'(u) = \frac{1}{u^2+1}$

$= 2\left[u\tan^{-1}u - \int \frac{u}{u^2+1}du\right] \surd\surd \qquad g'(u) = 1 \Rightarrow g(u) = u$

$= 2u\tan^{-1}u - \int \frac{2u}{u^2+1}du \surd$

(4)

2.4

$$\int y\,dx = \int\left(\frac{1}{\sin x} - \cos x\right)^2 dx$$

$$= \int\left(\frac{1}{\sin^2 x} - 2\frac{\cos x}{\sin x} + \cos^2 x\right)dx \qquad \surd\surd\surd$$

$$= \int\left(\cos ec^2 x - 2\cot x + \cos^2 x\right)dx \qquad \surd\surd$$

$$= -\cot x - 2\ln(\sin x) + \frac{x}{2} + \frac{\sin 2x}{4} + c \quad \surd\surd\surd$$

$$or\ -\cot x + 2\ln(\cos ecx) + \frac{x}{2} + \frac{\sin 2x}{4} + c \qquad \surd\surd\surd$$

or $\int\left[\frac{1-\sin x\cos x}{\sin x}\right]^2 dx$

$= \int \frac{(1-\sin x\cos x)^2}{\sin^2 x}dx$

$= \int \frac{1-2\sin x\cos x + \sin^2 x\cos^2 x}{\sin^2 x}dx$

$= \int \frac{1}{\sin^2 x} - \frac{2\sin x\cos x}{\sin^2 x} + \frac{\sin^2 x\cos^2 x}{\sin^2 x}dx$

$= \int (\cos ec^2 x - 2\cot x + \cos^2 x)dx$

(4)

2.5

$$\int y\,dx = \int \left(1+\cot^2 x\right)\cos ec^2 x\,dx$$

$$= \int \left(\cos ec^2 x + \cot^2 x \cos ec^2 x\right) dx \qquad \surd\surd$$

$$= -\cot x - \frac{\cot^3 x}{3} + c \qquad \surd\surd\surd\surd$$

Alternative

$$\int y\,dx = \int \left(1+\cot^2 x\right)(1+\cot^2 x)\,dx \qquad \surd$$

$$= \int (1+2\cot^2 x + \cot^4 x)dx$$

$$= \int (1+2\cot^2 x + \cot^2 x \cot^2 x)\,dx$$

$$= \int (1+2\cot^2 x + \cot^2 x(\cos ec^2 x - 1))\,dx \qquad \surd$$

$$= \int (1+2\cot^2 x + \cot^2 x \cos ec^2 x - \cot^2 x))\,dx$$

$$\int (1+\cot^2 x + \cot^2 x \cos ec^2 x)\,dx$$

$$= x - \cot x - x - \frac{1}{3}\cot^3 x + C \qquad \surd\surd\surd\surd$$

$$= -\cot x - \frac{\cot^3 x}{3} + c$$

(3)
[18]

QUESTION 3

3.1 $\dfrac{x-3}{\left[(x+2)(x-2)\right]^2} = \dfrac{x-3}{(x+2)^2(x-2)^2}\surd \qquad = \dfrac{A}{(x+2)} + \dfrac{B}{(x+2)^2} + \dfrac{C}{(x-2)} + \dfrac{D}{(x-2)^2} \qquad \surd$

$x-3 = A(x+2)(x-2)^2 + B(x-2)^2 + C(x-2)(x+2)^2 + D(x+2)^2 \ \surd \quad .....(1)$

When $x=2 \quad 2-3 = D(2+2)^2 \qquad D = -\dfrac{1}{16} \qquad \surd$

When $x=-2 \quad -2-3 = B(-2-2)^2 \quad B = -\dfrac{5}{16} \qquad \surd$

Equating $x^3 \quad A + C = 0 \quad \therefore \ C = -A \qquad \surd$

When $x=0 \quad -3 = A.2.4 + B.4 + C.-2.4 + D.4 \quad .....(2)\surd$

Substituting $D = -\dfrac{1}{16} \quad B = -\dfrac{5}{16} \quad C = -A \quad in\ (2)$

$-3 = 8A + \left(-\dfrac{5}{16}\right)4 + (-A)(-2.4) + \left(-\dfrac{1}{16}\right)4 \quad \therefore \ A = \dfrac{-3}{32} \ \surd \ C = \dfrac{3}{32}\surd$

$$\int \frac{x-3}{(x+2)^2(x-2)^2}dx = \int \frac{\frac{-3}{32}}{(x+2)}dx + \int \frac{-\frac{5}{16}}{(x+2)^2}dx + \int \frac{\frac{3}{32}}{(x-2)}dx + \int \frac{-\frac{1}{16}}{(x-2)^2}dx \surd$$

$$= \frac{-3}{32}\ln(x+2) + \frac{5}{16(x+2)} + \frac{3}{32}\ln(x-2) + \frac{1}{16(x-2)} + C \surd\surd\surd\surd$$

(7)

3.2 $\dfrac{x^2+x+1}{(x^2+1)(x-1)} = \dfrac{Ax+B}{x^2+1} + \dfrac{C}{x-1} \qquad \surd$

$x^2 + x + 1 = (Ax+B)(x-1) + C(x^2+1) \qquad \surd$

When $x=1 \quad 3 = 2C \quad \therefore \ C = \dfrac{3}{2} \quad \surd$

Expanding R.H.S $x^2 + x + 1 = Ax^2 + Bx - Ax - B + Cx^2 + C$

equating $x^2 \quad A + C = 1 \quad A + \dfrac{3}{2} = 1 \quad \therefore A = -\dfrac{1}{2} \qquad \surd$

equating $x \quad B - A = 1 \quad \therefore B = \dfrac{1}{2} \quad \surd$

$$\int \frac{x^2+x+1}{(x^2+1)(x-1)}dx = \int \frac{-\frac{1}{2}x + \frac{1}{2}}{x^2+1}dx + \int \frac{\frac{3}{2}}{x-1}dx \qquad \surd$$

$$= -\frac{1}{2}\int \frac{x}{x^2+1}dx + \frac{1}{2}\int \frac{1}{x^2+1}dx + \frac{3}{2}\int \frac{1}{x-1}dx \quad \surd$$

$$= -\frac{1}{4}\ln(x^2+1) + \frac{1}{2}\tan^{-1}x + \frac{3}{2}\ln(x-1) + C \quad \surd\surd\surd$$

(5)

[12]

QUESTION 4

4.1

$$x\frac{dy}{dx}-2y-3x^3+5x^2=0$$

$$\frac{dy}{dx}-\frac{2}{x}y=3x^2-5x \quad \surd$$

$$e^{\int pdx}=e^{\int -\frac{2}{x}dx}=e^{-2\ln x}=\frac{1}{x^2} \quad \surd\surd\surd$$

$$\int Qe^{\int pdx}dx=\int(3x^2-5x)\frac{1}{x^2}dx \surd\surd$$

$$=\int(3-\frac{5}{x})dx\surd$$

$$=3x-5\ln x\surd$$

$$y\frac{1}{x^2}=3x-5\ln x+C\surd\surd$$

$$x=2\ ;\ y=1\quad 1\left(\frac{1}{2^2}\right)=3(2)-5\ln 2+C \qquad \therefore C=-2,284\surd$$

Particular solution $y\frac{1}{x^2}=3x-5\ln x-2,284\surd$ (6)

4.2

$$\frac{d^2y}{dx^2}+2\frac{dy}{dx}+5y=10x^2$$

$$r^2+2r+5=0\surd$$

$$r=\frac{-2\pm\sqrt{4-20}}{2}=\frac{-2\pm\sqrt{-16}}{2}=\frac{-2\pm 4i}{2}=-1\pm 2i\surd\surd$$

$$y_c=e^{-x}\left[A\cos 2x+B\sin 2x\right]\surd$$

$$y=Cx^2+Dx+E \qquad \frac{dy}{dx}=2Cx+D \qquad \frac{d^2y}{dx^2}=2C\surd\surd\surd$$

$$2C+2(2Cx+D)+5(Cx^2+Dx+E)=10x^2$$

$$2C+4Cx+2D+5Cx^2+5Dx+5E=10x^2$$

Equating x^2 $\quad 5C=10 \qquad C=2\surd$

Equating x $\quad 4C+5D=0 \qquad 4(2)+5D=0 \qquad D=-\frac{8}{5}\surd$

Equating constants $\quad 2C+2D+5E=0 \qquad 4-\frac{16}{5}+5E=0 \qquad E=-\frac{4}{25}\surd$

$$y_p=2x^2-\frac{8}{5}x-\frac{4}{25}\surd$$

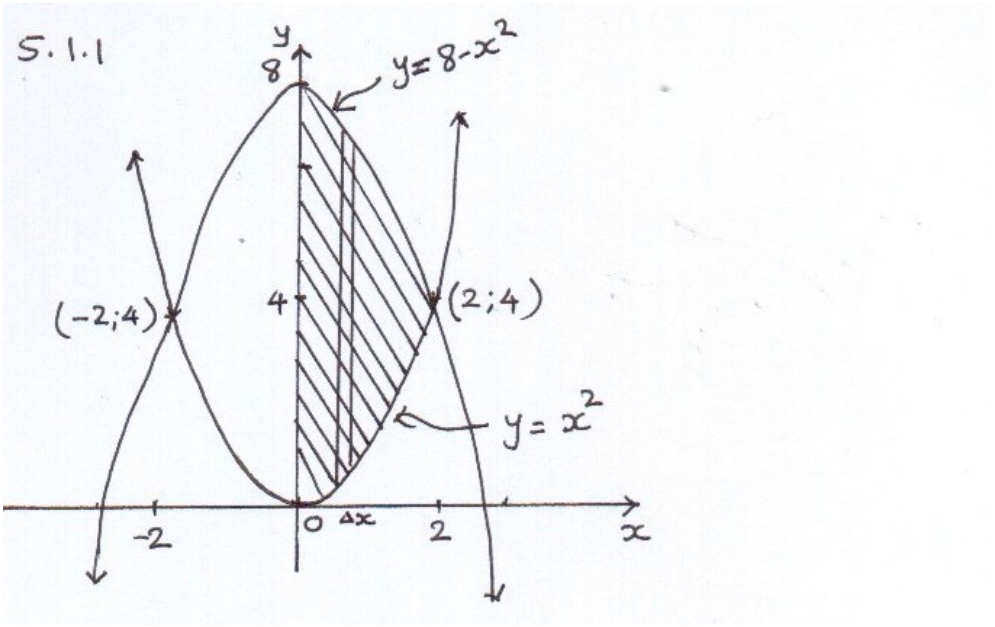
$$y=e^{-x}\left[A\cos 2x+B\sin 2x\right]+2x^2-\frac{8}{5}x-\frac{4}{25}\surd$$

(6)

[12]

QUESTION 5

5.1 5.1.1



√√ = each graph; √ = strip; √ = area

$y = x^2$

$y = 8 - x^2 \quad \therefore x^2 = 8 - x^2$

$2x^2 = 8 \qquad x = \pm 2 \qquad y = 4 \qquad (2;4) \qquad (-2;4)\surd\surd$ (3)

5.1.2

$$A = \int_a^b (y_1 - y_2)dx \surd$$

$$= \int_0^2 (8 - x^2 - x^2)dx \surd\surd$$

$$= \int_0^2 (8 - 2x^2)dx$$

$$= \left[8x - \frac{2}{3}x^3\right]_0^2 \surd$$

$$= \left[8(2) - \frac{2}{3}(2)^3 - (0)\right]_0^2 = \frac{32}{3} \text{ or } 10,667 units^2 \surd\surd$$

(3)

√ = Formula
√√ = Limits; function
√ = Integration
√ = Substitution of limits
√ = Answer

4 marks if area is negative

5.1.3

$$A_{m-x} = \int_a^b r\,dA \surd$$

$$= \int_a^b \frac{y_1 + y_2}{2}(y_1 - y_2)dx \surd$$

$$= \frac{1}{2}\int_a^b (y_1^2 - y_2^2\ )dx \ \surd \qquad or \quad 3ticks\ for\ this\ step$$

$$= \frac{1}{2}\int_0^2 \left\{(8 - x^2)^2 - \left(x^2\right)^2\right\} dx \surd\surd$$

$$= \frac{1}{2}\int_0^2 \left\{(64 - 16x^2 + x^4) - x^4\right\} dx \surd$$

$$= \frac{1}{2}\int_0^2 \left\{(64 - 16x^2\right\} dx \surd$$

$$= \frac{1}{2}\left[64x - 16\frac{x^3}{3}\right]_0^2 \surd$$

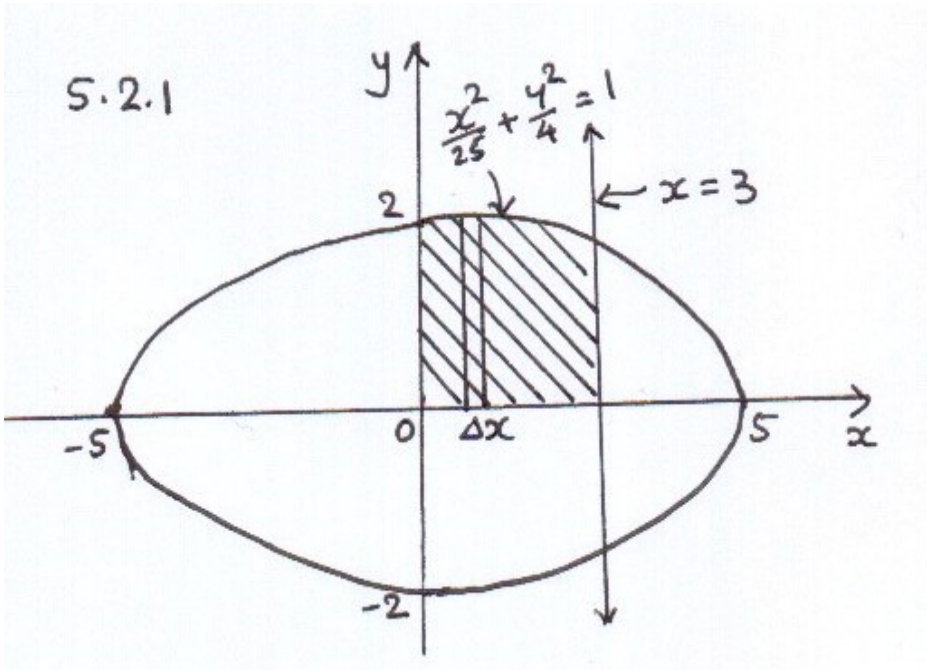
$$= \frac{1}{2}\left[64(2) - 16\frac{(2)^3}{3} - 0\right]\surd$$

$$= \frac{128}{3} \quad or \quad 42,667\ units^3 \surd$$

$$\overline{y} = \frac{A_{m-x}}{A} = \frac{\frac{128}{3}}{\frac{32}{3}} = 4 \quad or \frac{42,667}{10,667} = 4 \quad units \surd\surd$$

(6)

5.2 5.2.1



√√ = each graph; √ = strip; √ = area (2)

5.2.2 $V_x = \pi\int_a^b (y_1^2 - y_2^2\ )dx \quad or \quad \pi\int_a^b y^2 dx\ \surd$

$= \pi\int_0^3 \frac{4}{25}(25 - x^2\ )dx\ \surd\surd$

$= \frac{4}{25}\pi\left[25x - \frac{x^3}{3}\right]_0^3\ \surd$

$= \frac{4}{25}\pi\left[25(3) - \frac{(3)^3}{3} - 0\right]\ \surd$

$= 10,56\pi \quad or \quad 33,175\ units^3\ \surd$

(3)

5.2.3 $\bar{x} = \dfrac{V_{m-y}}{V_x}$

$V_{m-y} = \int_a^b r dV \surd = \int_a^b x\pi y^2 dx \surd$

$= \pi\int_0^3 x\dfrac{4}{25}(25-x^2)dx \surd\surd$

$= \dfrac{4}{25}\pi\int_0^3 (25x-x^3)dx \surd$

$= \dfrac{4}{25}\pi\left[25\dfrac{x^2}{2}-\dfrac{x^4}{4}\right]_0^3 \surd$

$= \dfrac{4}{25}\pi\left[25\dfrac{3^2}{2}-\dfrac{3^4}{4}-0\right]_0^3 = 14,76\pi \quad or 46,370 units^4 \surd\surd$

$\bar{x} = \dfrac{14,76\pi}{10,56\pi} \quad or \quad \dfrac{46,370}{33,175} = 1,398 units \surd\surd$

Or $V_{m-y} = \int_a^b r\,dv$

$= \pi\int_a^b x(y_1^2 - ¿\, y_2^2)\,dx \quad ¿\surd\surd$

$= \pi\int_0^3 4x\left(1-\dfrac{x^2}{25}\right)dx \quad \surd\surd$

$= 4\pi\int_0^3 \left(x-\dfrac{x^3}{25}\right)dx \quad \surd$

¿ $4\pi\left[\dfrac{x^2}{2}-\dfrac{x^4}{100}\right]_0^3 \surd$

¿ $4\pi\left[\dfrac{9}{2}-\dfrac{3^4}{100}-0\right] \surd$

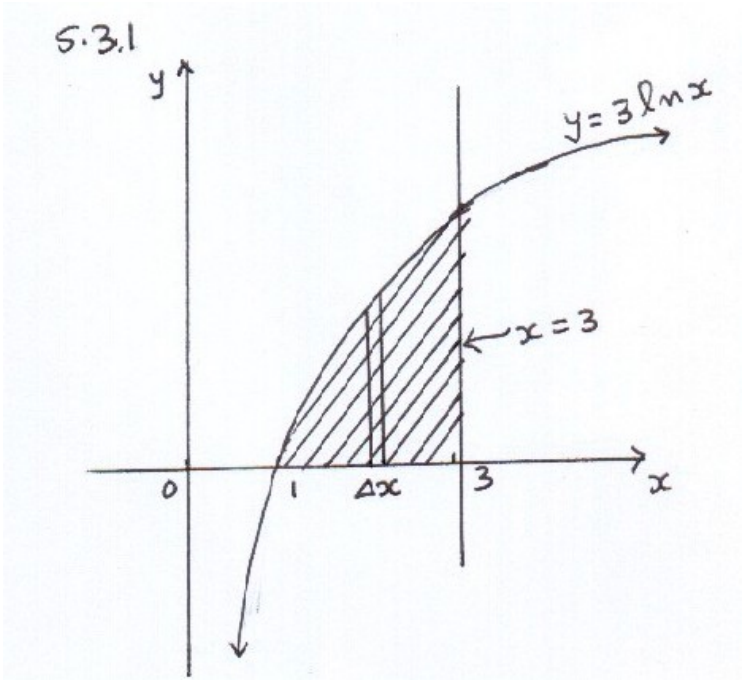
¿ $\dfrac{369}{25}\pi \surd$

$= 14,76\pi$

$= 46,370\, units^3$

(5)

5.3.1



√√ = each graph; √ = *x*-intercept of y = 3lnx; √ = area (No marks may be allocated for the strip, as it was given in the question.) (2)

5.3.2

$$A = \int_a^b (y_1 - y_2)dx \qquad or \quad \int_a^b y dx \surd$$

$$= \int_1^3 3\ln x\, dx \surd\surd$$

$$= 3\left[x\ln x - x\right]_1^3 \surd$$

$$= 3\left[3\ln 3 - 3 - (1\ln 1 - 1)\right] \surd$$

$$= 3\left[3\ln 3 - 2\right] = 3,888 units^2 \surd$$

(3)

5.3.3

$$I_y = \int_a^b r^2\,dA \surd$$

$$= \int_a^b x^2(y_1 - y_2)\,dx \quad or \quad \int_a^b x^2 y\,dx \surd$$

$$= \int_1^3 x^2 3\ln x\,dx \surd\surd$$

$$= 3\left[\ln x\frac{x^3}{3} - \int\frac{1}{x}\frac{x^3}{3}dx\right]_1^3 \qquad \surd\surd$$

$$= 3\left[\ln x\frac{x^3}{3} - \frac{1}{3}\int x^2 dx\right]_1^3 \qquad \surd$$

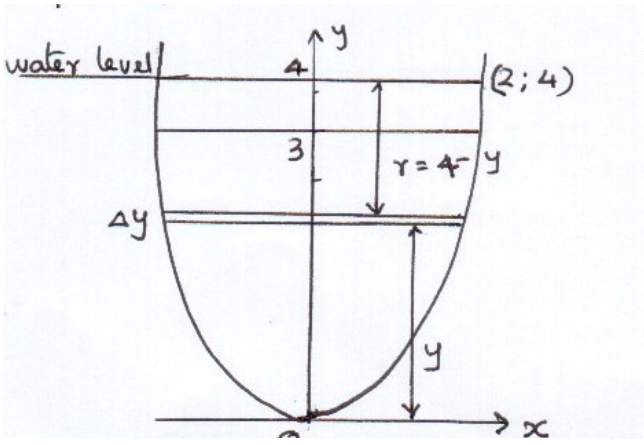
$$= 3\left[\ln x\frac{x^3}{3} - \frac{1}{3}\frac{x^3}{3}\right]_1^3 \quad \surd$$

$$= 3\left[\ln 3\frac{3^3}{3} - \frac{1}{3}\frac{3^3}{3} - \left(\ln 1\frac{1^3}{3} - \frac{1}{3}\frac{1^3}{3}\right)\right] \quad \surd$$

$$= 20,996 units^4 \qquad \surd$$

(5)

5.4 5.4.1 ***x*-axis at the bottom**



$y = ax^2$

$4 = a(2)^2 \surd \quad \therefore \quad a = 1 \qquad \therefore y = x^2 \quad \surd \; or \quad x = y^{\frac{1}{2}}$

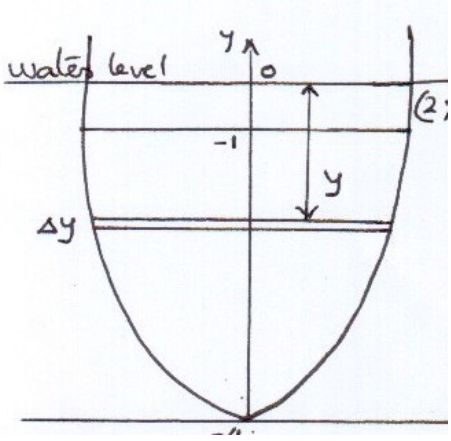
$y = ax^2 - 4$

$(2;0) \qquad 0 = a(2)^2 - 4 \quad \surd \qquad \therefore a = 1$

$\therefore y = x^2 - 4 \surd \quad or \quad x = (y+4)^{\frac{1}{2}}$ (6)

√ = water level at 4; √ = height of wall at 3; √ = point (2;4); √ = strip

***x*-axis at the top**



√ = *x*-axis at water level; height of wall at –1
√ = point (2;0)
√ = strip

(3)

5.4.2

Second moment of area $= \int_a^b r^2 dA \surd$

$= \int_0^3 (4-y)^2 2y^{\frac{1}{2}} dy \surd\surd$

$= 2\int_0^3 (16-8y+y^2) y^{\frac{1}{2}} dy \surd$

$= 2\int_0^3 (16y^{\frac{1}{2}} - 8y^{\frac{3}{2}} y + y^{\frac{5}{2}}) dy \surd$

$= 2\left[16\frac{y^{\frac{3}{2}}}{\frac{3}{2}} - 8\frac{y^{\frac{5}{2}}}{\frac{5}{2}} + \frac{y^{\frac{7}{2}}}{\frac{7}{2}} \right]_0^3 \surd$

$= 2\left[\frac{32}{3} y^{\frac{3}{2}} - \frac{16}{5} y^{\frac{5}{2}} + \frac{2}{7} y^{\frac{7}{2}} \right]_0^3$

$= 2\left[\frac{32}{3}(3)^{\frac{3}{2}} - \frac{16}{5}(3)^{\frac{5}{2}} + \frac{2}{7}(3)^{\frac{7}{2}} - 0 \right] \surd$

$= 37,808 m^4 \surd$

$\bar{y} = \frac{37,808}{15,242} = 2,481\ m \surd\surd$

(10)

second moment of area $= \int_a^b r^2 dA$

$= \int_{-4}^{-1} y^2 2(y+4)^{\frac{1}{2}} dy$

$u = y+4 \quad dy = du \quad y = u-4$

$y = -4 \quad u = 0 \quad \text{and} \quad y = -1 \quad u = 3$

$= 2\int_0^3 (u-4)^2 u^{\frac{1}{2}} du$

$= 2\int_0^3 (u^2 - 8u + 16) u^{\frac{1}{2}} du$

$= 2\int_0^3 \left(u^{\frac{5}{2}} - 8u^{\frac{3}{2}} + 16u^{\frac{1}{2}} \right) du$

$= 2\left[\frac{2}{7} u^{\frac{7}{2}} - 8\left(\frac{2}{5}\right) u^{\frac{5}{2}} + 16\left(\frac{2}{3}\right) u^{\frac{3}{2}} \right]_0^3$

$= 2\left[\frac{2}{7} 3^{\frac{7}{2}} - 8\left(\frac{2}{5}\right) 3^{\frac{5}{2}} + 16\left(\frac{2}{3}\right) 3^{\frac{3}{2}} - 0 \right]$

$= 37,808\ m^4$

$\bar{y} = \frac{37,808}{-15,242} m = -2,481 m \surd$

(5)

[40]

QUESTION 6

6.1

$$S=\int_a^b \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx \surd$$

$$y^{\frac{2}{3}}=4-x^{\frac{2}{3}}$$

$$y=\left(4-x^{\frac{2}{3}}\right)^{\frac{3}{2}} \surd$$

$$\frac{dy}{dx}=\frac{3}{2}\left(4-x^{\frac{2}{3}}\right)^{\frac{1}{2}}.-\frac{2}{3}x^{-\frac{1}{3}} \surd\surd$$

$$=-\frac{\left(4-x^{\frac{2}{3}}\right)^{\frac{1}{2}}}{x^{\frac{1}{3}}}$$

$$\left(\frac{dy}{dx}\right)^2=\frac{\left(4-x^{\frac{2}{3}}\right)}{x^{\frac{2}{3}}} \surd$$

$$1+\left(\frac{dy}{dx}\right)^2=1+\frac{4-x^{\frac{2}{3}}}{x^{\frac{2}{3}}}=\frac{x^{\frac{2}{3}}+4-x^{\frac{2}{3}}}{x^{\frac{2}{3}}}=\frac{4}{x^{\frac{2}{3}}} \surd$$

$$S=\int_0^4 \sqrt{\frac{4}{x^{\frac{2}{3}}}}dx \surd$$

$$=\int_0^4 \frac{2}{x^{\frac{1}{3}}}dx=2\int_0^4 x^{-\frac{1}{3}} \surd\surd$$

$$=2\left[\frac{3}{2}x^{\frac{2}{3}}\right]_0^4 \surd=2\left[\frac{3}{2}4^{\frac{2}{3}}-0\right]\surd=7,560\surd$$

Alternative

$$S=\int_a^b \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx \surd$$

$$y^{\frac{2}{3}}=4-x^{\frac{2}{3}}$$

$$\frac{2}{3}y^{-\frac{1}{3}}\frac{dy}{dx}=-\frac{2}{3}x^{-\frac{1}{3}} \surd\surd$$

$$\frac{dy}{dx}=\frac{-\frac{2}{3}x^{-\frac{1}{3}}}{\frac{2}{3}y^{-\frac{1}{3}}} \surd$$

$$=-\frac{y^{\frac{1}{3}}}{x^{\frac{1}{3}}} \surd$$

$$1+\left(\frac{dy}{dx}\right)^2=1+\frac{y^{\frac{2}{3}}}{x^{\frac{2}{3}}}=\frac{x^{\frac{2}{3}}+y^{\frac{2}{3}}}{x^{\frac{2}{3}}} \surd$$

$$=\frac{4}{x^{\frac{2}{3}}} \surd \text{ because}(x^{\frac{2}{3}}+y^{\frac{2}{3}}=4 \text{ ,the curve given })$$

$$S=\int_0^4 \sqrt{\frac{4}{x^{\frac{2}{3}}}}dx \surd$$

$$=\int_0^4 \frac{2}{x^{\frac{1}{3}}}dx=2\int_0^4 x^{-\frac{1}{3}}dx \surd$$

$$=2\left[\frac{3}{2}x^{\frac{2}{3}}\right]_0^4=2\left[\frac{3}{2}4^{\frac{2}{3}}-0\right]=7,560\surd\surd\surd$$

(6)

6.2

$x = 3t^2 + 6t \qquad \frac{dx}{dt} = 6t + 6\ \surd \qquad \left(\frac{dx}{dt}\right)^2 = (6t+6)^2 = 36t^2 + 72t + 36$

$y = 3t + 3 \qquad \frac{dy}{dt} = 3\ \surd \qquad \left(\frac{dy}{dt}\right)^2 = 9$

$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 36t^2 + 72t + 36 + 9 = 36t^2 + 72t + 45\ \surd \qquad = 9(4t^2 + 8t + 5)$

$A_x = 2\pi \int_{t_1}^{t_2} y \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt\ \surd$

$= 2\pi \int_0^1 (3t+3)\sqrt{9(4t^2+8t+5)}\, dt \quad \surd\surd \qquad \lim its;\ for\ substituting\ y = 3t+3$

$= 2\pi \int_0^1 3(t+1)3\sqrt{(4t^2+8t+5)}\, dt\ \surd$

$= 2\pi \int_5^{17} 3(t+1)3u^{\frac{1}{2}} \frac{du}{8(t+1)}\ \surd\surd$

$= 2\pi \frac{9}{8} \int_5^{17} u^{\frac{1}{2}}\, du = \frac{9}{4}\pi \left[\frac{2}{3}u^{\frac{3}{2}}\right]_5^{17}\ \surd$

$= \frac{3}{2}\pi \left[17^{\frac{3}{2}} - 5^{\frac{3}{2}}\right]\ \surd = 277,618 units^2\ \surd$

$u = 4t^2 + 8t + 5$

$\frac{du}{dt} = 8t + 8$

$dt = \frac{du}{8(t+1)} \qquad t = 0 \quad u = 5 \quad and$

$t = 1 \quad u = 4 + 8 + 5 = 17$

Alternative

$$A_x = 2\pi \int_{t_1}^{t_2} y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt$$

$$= 2\pi \int_0^1 (3t+3)\sqrt{(36t^2+72t+45)}\,dt$$

$u = 36t^2 + 72t + 45$

$\frac{du}{dt} = 72t + 72 \qquad dt = \frac{du}{72t+72}$

$t = 0 \qquad u = 45$

$t = 1 \qquad u = 153$

$$= 2\pi \int_{45}^{153} (3t+3)u^{\frac{1}{2}} \frac{du}{72t+72}$$

$$= 2\pi \frac{3}{72} \int_{45}^{153} u^{\frac{1}{2}} du$$

$$2\pi \frac{3}{72}\frac{2}{3}\left[u^{\frac{3}{2}}\right]_{45}^{153}$$

$$\frac{\pi}{18}\left[153^{\frac{3}{2}} - 45^{\frac{3}{2}}\right]$$

$$= 277,618 \text{units}^2$$

(6)
[12]

TOTAL: 100

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